

How to order this LED panel

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Generic instructions for JLCPCB. Screens change; pick the same meaning, not the same pixel.

Three things people get wrong

Use LeadFree HASL (not leaded). Set inner copper to 1 oz (not the 0.5 oz default). Turn on Confirm parts placement for the first batch. A rotated SK6812 will break the daisy chain.

01 Files to upload

Use the three files that sit next to the KiCad project file (.kicad_pro). Ignore any extra gerbers/ folder.

Setting	Select this	Why
<name>_gerbers.zip	PCB / Gerber upload	Copper, mask, silk, drills. One zip only.
<name>_bom.csv	BOM	LCSC part numbers for assembly.
<name>_cpl.csv	CPL / centroid	Pick-and-place positions, top side.

Open Gerber Viewer after upload. You want a circular 4-layer board, parts on the top only, DFM all green. Red DFM: stop and re-export before you pay.

02 Board type

Setting	Select this	Why
Base material	FR-4	Normal fibreglass. Aluminium / copper-core is a different heatsink design.
Layers	4	F.Cu data+GND, In1.Cu 5V, In2.Cu GND, B.Cu GND. Not 2, not 6+.
PCB qty	5 (or more)	Five is the usual first run. More copies drop the unit price.
Product type	Industrial / Consumer	Lighting panel. Medical / aerospace only adds paperwork.
Different design	1	One Gerber set. Raise only if several designs are in one zip.
Delivery format	Single PCB	Finished circles. Rails for SMT are added later, then cut off.
PCB thickness	1.6 mm	Stiff enough for M3 holes. Thinner 4-layer stock warps in reflow.

03 Look and laminate

Setting	Select this	Why
PCB colour	Green	Cheapest, best process control. Black is optional cosmetics.
Silkscreen	White	Readable on green. 5V / GND / DATA are board labels and must stay.
Material type	CHANGE: FR4-TG155	Lead-free reflow is hot. TG155 has more margin than TG135.
Surface finish	CHANGE: LeadFree HASL	RoHS, matches high-temp paste. Leaded HASL is not RoHS. ENIG is optional luxury.

04 Copper and vias

Setting	Select this	Why
Outer copper	1 oz	Enough on a full plane. 2 oz is extra cost.
Inner copper	CHANGE: 1 oz	5V lives on an inner plane (~7 A). Do not leave 0.5 oz.
Layer sequence	No	Standard 4-layer order already matches the design.
Specify stackup	No	No impedance control. SK6812 is not USB.
Via covering	Plugged	Vias sit next to capacitors. Mask keeps solder out of the hole.
Via plating	Horizontal electroless Cu	Normal 4-layer plating. Leave as offered.
Min via hole	0.3 mm / 0.4-0.45 mm	Matches the design. Smaller vias cost more and are unused.
Outline tolerance	+/- 0.2 mm Regular	A 157 mm circle does not need precision outline.

05 Marks and tests

Setting	Select this	Why
Confirm production file	Yes (first order)	A quick human check of the Gerbers. Cheap insurance once.
Mark on PCB	Remove Mark	Hides the order number. Cleaner photos.
Electrical test	Flying probe fully test	Normal net test on a 4-layer board.
Gold fingers, slots, UL	All No	Castellated, press-fit, edge plate, backdrill, humidity card: unused.

06 Assembly

Switch PCB Assembly On. Turn it off only if you want bare boards and will skip this section.

Setting	Select this	Why
Assemble	Top side only	Every SMD part is on F.Cu. The back is a GND plane.
PCBA type	Standard	Economic rejects more parts. Power caps are polar.
PCBA qty	Same as PCB qty	Usually 5.
Edge rails / fiducials	Added by JLCPCB	A round board cannot run SMT without rails. Size bump is temporary.
Confirm parts placement	CHANGE: Yes (first batch)	One rotated SK6812 kills a daisy. Check the photo, then they run the rest.
Stencil / fixture storage	No	Not a weekly repeat order.
Parts selection	By customer (self-service)	BOM already has LCSC codes. Use Manual Match only if a line will not map.

07 Assembly extras

Setting	Select this	Why
Bake / photo / clean / coat	No	Indoor panel, no BGA moisture parts, no conformal coat.
Special stencil	No	Standard stencil covers 0603 and LED pads.
Depanel + remove rails	Yes	They added rails for SMT. This returns the circular board.
Assembly flying probe	No	Bare PCB already tested. No in-house function jig.
Function test	No	You power it: 5V, GND, DATA.
Packaging	Antistatic bubble film	Enough for assembled boards.
Solder paste	High temp.	Lead-free process. Required with LeadFree HASL.
Nitrogen reflow	No	Cosmetic only.
Extra stencil product	Off	That is for hand assembly at home. They already have one.

08 Parts cart

Every BOM line must be In Stock. Typical mapping:

Setting	Select this	Why
C5348912	SK6812 / SKC6812 x109	Do not swap to a different LED footprint.
C14663	100 nF 0603 x109	Local decoupling.
C79111	220 uF 6.3 V x2	Polar polymer. Plus toward 5V.
C7484	SN74AHCT1G125 x1	5 V buffer, OE tied to GND.
C502527	SD05 TVS x1	On DATA_IN.
C23138	330 ohm 0603 x1	Series on data into the first LED.

Wire pads and M3 holes are not assembled. You solder wires yourself. If a part is out of stock, an alternate is OK only when package and voltage match.

09 Before you pay

- ☐ 4 layers, 1.6 mm, FR-4 TG155, green, white silk
- ☐ LeadFree HASL, not HASL (with lead)
- ☐ Outer 1 oz and inner 1 oz, not 0.5 oz inner
- ☐ Vias plugged, min drill 0.3 mm
- ☐ Assembly on, top side, Standard, rails by JLCPCB, depanel Yes
- ☐ Confirm parts placement Yes (first order)
- ☐ High-temp paste, no extra stencil, no coating
- ☐ Latest <name>_gerbers.zip + _bom.csv + _cpl.csv, DFM green

When it arrives

One fat 5 V cable to the 5V and GND holes. Data to the DATA hole. Do not run LED current through an ESP32 DevKit. Tap 5V/GND at the panel.